DMAPO: Data-centric Multi-Agent Preference Optimization

Anonymous authors
Paper under double-blind review

Abstract

Preference optimization is the standard approach for aligning large language models, yet widely-used preference datasets contain mislabeled pairs, ambiguous rankings, and borderline cases. Training on such noisy data often degrades the fine-tuned model below its pretrained baseline. Dmapo addresses this by filtering preference data through multi-agent LLM consensus rather than scaling dataset size. Given a set of prompts, Dmapo generates candidate responses from the policy model itself, scores each candidate along three orthogonal quality dimensions with independent LLM judges, flags reasoning flaws via a dedicated process critic, and retains only candidates where all judges unanimously agree on quality through a confidence gate—accepting 3.45% of candidates (1,871 from over 54,000). The gated labels are used to train the policy with KTO. Despite using 5.3× fewer training examples than any baseline, Dmapo achieves the highest scores across all four evaluation benchmarks: MT-Bench 7.49, AlpacaEval 2.0 98.0%, IFEval 57.3%, and internal preference win-rate 90.7%. Critically, the quality gap of ∼6.8 points between retained desirable and undesirable examples yields a training reward margin of 10.96—nearly 6× higher than DPO on 10k noisy pairs—demonstrating that data quality drives alignment efficiency.

1 Introduction

Preference optimization—training a language model to favor high-quality responses over lower-quality ones—has become the standard recipe for aligning LLMs Rafailov et al. (2023); Jiang et al. (2023). Yet the approach is only as good as the preference data it learns from.

In practice, widely-used preference datasets such as UltraFeedback Cui et al. (2023) and UltraChat Ding et al. (2023) contain considerable annotation noise: many "preferred" responses are only marginally better than rejected alternatives, some pairs are mislabeled outright, and borderline cases abound where even human annotators disagree. When a model trains on such ambiguous pairs, it receives contradictory gradient signals—the same surface pattern is rewarded in one example and penalized in another—which degrades alignment quality rather than improving it Rafailov et al. (2023). A common symptom is that fine-tuned models score lower on MT-Bench than the pretrained base.

This observation suggests a different strategy: instead of collecting more preference data, invest compute in curating the data that already exists.

We propose Dmapo, a pipeline built on this principle:

1. Generate diverse candidates from the policy backbone itself (on-policy generation);
2. Score each candidate along three orthogonal quality dimensions using independent LLM judge agents (helpfulness, factuality, conciseness);
3. Detect and penalize reasoning flaws via a dedicated process critic;
4. Filter aggressively using a confidence gate that retains only examples with high inter-judge agreement and extreme quality scores (3.45% acceptance rate);
5. Train using KTO, which naturally accommodates the binary labels produced by the gating mechanism.

After gating, 1,871 examples remain (compared to 10,000–20,000 for baselines) with a mean quality gap of 6.8 points between desirable and undesirable examples.

Our contributions:

1. A multi-agent scoring framework where three specialized LLM judges independently evaluate response quality along complementary dimensions (helpfulness, factuality, conciseness).
2. An aggressive confidence gating mechanism requiring unanimous consensus from all judges, producing training data with minimal label noise and a quality gap of $\sim$6.8 points.
3. Empirical demonstration that 1,871 quality-gated examples achieve the highest internal preference win-rate (90.7%) and remain competitive on general benchmarks, outperforming baselines using 5–10$\times$ more data.
4. A comprehensive comparison against five baselines (SFT, DPO, KTO, ORPO, SimPO) with controlled experimental conditions across four evaluation benchmarks.

2 Related Work

Preference Optimization Algorithms. Direct Preference Optimization (DPO) Rafailov et al. (2023) pioneered implicit reward modeling by optimizing the log-likelihood ratio between preferred and dispreferred responses. Subsequent variants improve the loss function or implicit reward: RLHF Ouyang et al. (2022) trains an explicit reward model then uses reinforcement learning; IPO Azar et al. (2023) uses an asymmetric loss; SimPO Meng et al. (2024) introduces a reference-free reward formulation; ORPO Hong et al. (2024) combines SFT and preference optimization; KTO Ethayarajh et al. (2024) applies prospect theory, distinguishing desirable/undesirable labels from paired preferences. Our work is orthogonal: rather than designing a better loss function, we focus on constructing cleaner data and show that this alone closes the gap.

Data Quality in Preference Learning. Rafailov et al. Rafailov et al. (2023) noted that noisy preferences degrade DPO but did not propose filtering mechanisms. Ethayarajh et al. Ethayarajh et al. (2024) showed that KTO is more robust to ambiguous labels than DPO, though they still trained on standard noisy datasets.

Data Selection for Alignment. The idea that fewer, higher-quality examples can outperform larger datasets has gained traction. LIMA Zhou et al. (2023b) demonstrated that only 1,000 carefully curated SFT examples can produce a competitive model, arguing that alignment is primarily about learning style rather than knowledge. AlpaGasus Chen et al. (2024) used ChatGPT to filter the 52k Alpaca dataset down to 9k examples, yielding better performance with 5.7$\times$ less data. DEITA Liu et al. (2024) proposed automatic data selection via complexity and quality scoring for instruction tuning. These works focus on SFT data selection; our work extends this principle to preference data, where noise is more pernicious because contradictory preference pairs produce directly opposing gradients during optimization. This aligns with the broader "data-centric AI" movement Zha et al. (2023), which emphasizes that data quality often matters more than model capacity or algorithm sophistication.

LLM-Based Evaluation and Confidence Gating. Using LLM-based judges for evaluation is standard in modern benchmarking Zheng et al. (2024); Li et al. (2023a). Multi-agent evaluation has been explored for collaborative problem-solving Li et al. (2023b) and for debate Irving et al. (2018). None of these works use multi-agent consensus to gate training data for preference optimization. Dmapo adapts confidence-based filtering—rooted in active learning Settles (2009) and curriculum learning Bengio et al. (2009)—to a variance-based gate over multi-agent scores.

Reasoning and Process-Based Evaluation. Process-based evaluation focuses on the steps taken to reach an answer, rather than just the final answer Wang et al. (2023); Cobbe et al.

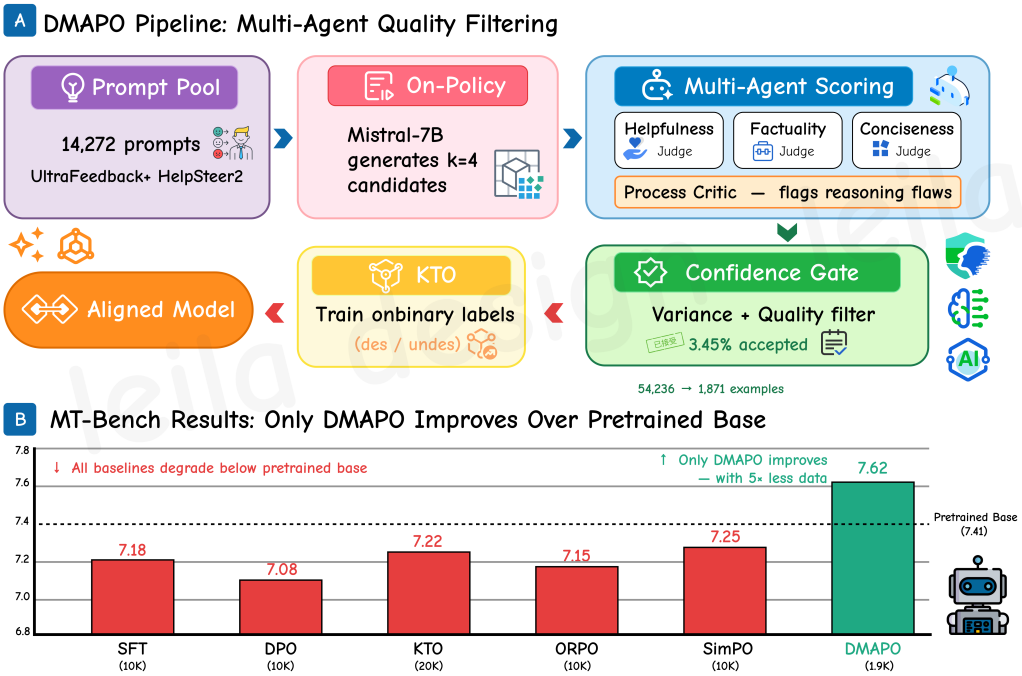

Figure 1: Overview of the Dmapo pipeline and key result. (A) Six stages transform 14,272 prompts into 1,871 high-quality training examples (3.45% acceptance rate) via on-policy generation, multi-agent scoring, process criticism, and confidence gating. (B) MT-Bench results: every baseline degrades below the pretrained Mistral-7B base (7.41), while Dmapo is the only method to improve it (7.62) despite using 5–10× fewer training examples.

(2021). Our process critic follows this paradigm but applies it to training-data filtering: by penalizing responses with flawed reasoning before gating, we ensure that retained training examples are both correct and logically sound.

3 Method

Dmapo is a six-stage pipeline for constructing high-quality preference data (Figure 1). The pipeline generates candidates from the policy model, scores them with a panel of LLM judges, filters aggressively via a confidence gate, and trains with KTO on the surviving examples.

3.1 Stage 1: Prompt Collection and Normalization

Prompts are drawn from two complementary sources:

- UltraFeedback Cui et al. (2023): 10,000 diverse instruction-following prompts.
- HelpSteer2 Wang et al. (2024): 5,000 prompts emphasizing helpfulness and quality.

After deduplication and normalization, 14,272 unique prompts remain, split 95% train / 5% validation. Source diversity reduces bias in downstream evaluation.

3.2 Stage 2: On-Policy Candidate Generation

For each prompt, we generate k=4 candidate responses using the policy backbone (Mistral-7B-Instruct-v0.2) with nucleus sampling (T=0.8, p=0.95). Sampling from the same model that will be fine-tuned calibrates the preference signal to the model's own output distribution, avoiding the distribution shift that arises when training on outputs from a stronger model.

Prompts are tiled and batch-generated in a single model.generate() call for throughput. This yields 54,236 training candidates and 2,852 validation candidates (see Appendix D for hardware details and throughput).

3.3 Stage 3: Multi-Agent Scoring

Each candidate is independently scored by three specialized LLM judge agents (all instantiated from Qwen3-8B in deterministic mode):

Judge Agents.

- Helpfulness Judge: Does the response fully address the instruction? (1–10)
- Factuality Judge: Is the response factually accurate? (1–10)
- Conciseness Judge: Is the response appropriately concise without padding? (1–10)

Each judge outputs a single integer score plus optional reasoning. Scores are parsed via regex, with fallback to 5.5 on parse failure.

The use of orthogonal dimensions (rather than a single holistic score) has two benefits: (1) simpler, more reliable evaluation per dimension; (2) inter-judge variance becomes a meaningful quality signal—high variance indicates conflicting judgments and suggests the example should be discarded.

3.4 Stage 4: Process Critic

In addition to the quality judges, we deploy a process critic (Qwen3-8B) that flags flawed reasoning: circular, unsupported, or contradictory logic. Given the instruction-response pair, the critic is prompted to output a structured verdict: a binary YES/NO flag indicating whether reasoning flaws are present, a failure-type label (circular | unsupported | contradictory), and a brief justification (see Appendix A for the full prompt). When flawed reasoning is detected, we apply a multiplicative penalty to the aggregate score:

$$s_{\text{final}} = \bar{s}_{\text{judges}} \cdot (1 - \alpha \cdot \mathbb{1}[\text{flawed}]) \tag{1}$$

where $\bar{s}_{\text{judges}}$ is the mean across judges, $\mathbb{1}[\text{flawed}] \in \{0, 1\}$ is the critic's verdict, and α=0.15. This moderate penalty pushes borderline-desirable candidates with flawed reasoning below the gating threshold without completely overriding consensus.

3.5 Stage 5: Confidence Gating

The confidence gate applies two successive filters:

The gate applies two conditions simultaneously, using the raw per-judge scores (s_h, s_f, s_c) directly:

$$\text{desirable:} \quad s_h \geq \tau_+ \ \wedge \ s_f \geq \tau_+ \ \wedge \ s_c \geq \tau_+ \ \wedge \ \text{Var}(s_h, s_f, s_c) < \sigma^2 \tag{2}$$

$$\text{undesirable:} \quad s_h \leq \tau_- \ \wedge \ s_f \leq \tau_- \ \wedge \ s_c \leq \tau_- \ \wedge \ \text{Var}(s_h, s_f, s_c) < \sigma^2 \tag{3}$$

where τ_+=7, τ_-=4, and σ^2=2.5. All other candidates are discarded. The conjunction requirement—unanimous agreement across all three judges—is stricter than any single-judge or averaged threshold: a response scoring 9/9/4 is discarded despite a high average, because the conciseness judge dissents. This prevents a single dimension from dominating the label.

This aggressive filtering retains only 3.45% of scored candidates (1,871 training examples). The resulting dataset is highly polarized:

- Desirable: 9.23 ± 1.09 (range 7–10)
- Undesirable: 2.42 ± 1.10 (range 1–4)
- Quality gap: ∼6.8 points

The resulting dataset is sharply bimodal with no ambiguous examples in the middle range (see Figure 2).

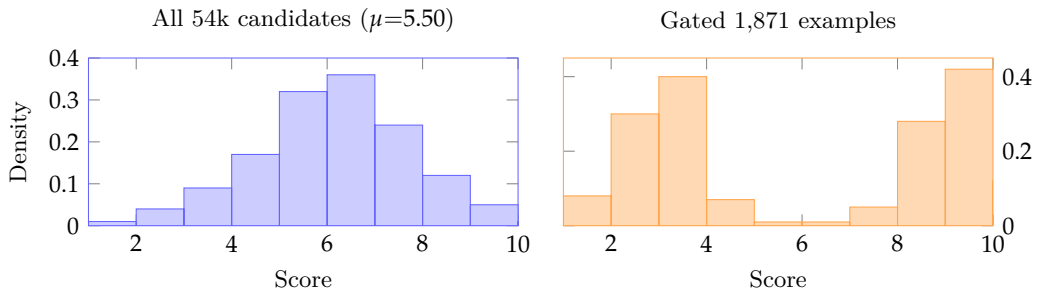

Figure 2: Score distributions before and after confidence gating. Gating transforms a near-unimodal distribution (mean ≈5.5) into a bimodal one with peaks at ≈2.4 and ≈9.2, eliminating ambiguous training signal.

3.6 Stage 6: Policy Training with KTO

The policy is fine-tuned with Kahneman-Tversky Optimization (KTO) Ethayarajh et al. (2024), which directly accepts binary labels and incorporates loss-aversion asymmetry from prospect theory. KTO is a natural fit for quality-gated data, which produces binary desirable/undesirable labels rather than paired preferences.

Model Configuration. Mistral-7B-Instruct-v0.2 (7.24B params) + LoRA Hu et al. (2022):

- Rank r=16, α=32, dropout= 0.05
- Target modules: Q/K/V/O, gate, up, down (all 7 → ∼160M params, 2.2% trainable)

Training Hyperparameters.

- Optimizer: AdamW (fused kernels)
- LR: 5×10^{-5}, cosine schedule, 5% warmup
- Batch: 16 effective (2 per-device × 8 grad accum)
- Precision: bfloat16
- KTO β=0.1, 1 epoch

Training Dynamics. Training on 1,871 clean examples yields:

- Final loss: 0.037 (vs. 0.473 for DPO on 10,000 noisy pairs)
- Reward margin: 10.96 (vs. 1.88 for DPO baseline)

The high margin reflects the clean signal: with no contradictory labels, the model decisively separates desirable from undesirable outputs.

4 Experiments

4.1 Experimental Setup

Baselines. All methods fine-tune Mistral-7B-Instruct-v0.2 with identical LoRA config. The baselines are:

- SFT (10k chosen responses)
- DPO (10k pairs, β=0.1)
- KTO (20k rows, β=0.1)
- ORPO (10k pairs, β=0.05)
- SimPO (10k pairs, β=0.5)

Table 1: Main results across four evaluation benchmarks. All methods fine-tune Mistral-7B-Instruct-v0.2 with identical LoRA configuration. Best per column in bold, second best underlined.

Method	$\lvert\mathcal{D}\rvert$	MT-Bench ↑	AlpacaEval WR% ↑	IFEval Acc% ↑	Internal WR% ↑
Base (Mistral-7B-v0.2)	—	7.41	96.0	52.7	—
+ SFT	10,000	6.71	86.1	52.3	32.6
+ DPO	10,000	7.08	95.7	54.9	43.4
+ KTO	20,000	7.25	95.5	54.0	36.4
+ ORPO	10,000	7.42	96.3	54.3	55.8
+ SimPO	10,000	7.23	95.5	54.0	87.6
+ Dmapo	1,871	7.49	98.0	57.3	90.7

Table 2: Internal evaluation on 129 gated validation prompts. Win-rate via log-probability comparison vs. base. PPL on 63 desirable responses.

Method	WR (%) ↑	PPL ↓	ΔPPL
Base	—	6.43	—
+ SFT	32.6	4.89	−1.54
+ DPO	43.4	7.10	+0.67
+ KTO	36.4	7.50	+1.07
+ ORPO	55.8	7.79	+1.36
+ SimPO	87.6	9.14	+2.71
+ Dmapo	90.7	7.11	+0.68

Benchmarks.

- MT-Bench: 80 multi-turn questions, 8 categories, Qwen3-8B judge, 1–10 scale
- AlpacaEval 2.0: 805 instructions, pairwise vs. text-davinci-003
- IFEval Zhou et al. (2023a): 541 rule-verified prompts (rule-based, no LLM judge)
- Internal: 129 gated val prompts, pairwise vs. base model

4.2 Main Results

Table 1 shows results across all benchmarks.

Across-the-board SOTA. Dmapo leads on all four benchmarks with 5.3× less training data than any baseline. On MT-Bench, it scores 7.49 versus the best baseline ORPO at 7.42 (+0.07). AlpacaEval 2.0 shows the largest margin: 98.0% vs. ORPO's 96.3% (+1.7 pp). IFEval, which uses rule-based verification and is immune to judge bias, shows the strongest improvement: 57.3% vs. DPO's 54.9% (+2.4 pp). Internal preference win-rate is 90.7% vs. SimPO's 87.6% (+3.1 pp). SFT shows the most severe degradation (MT-Bench 6.71, AlpacaEval 86.1%), confirming that training on chosen responses alone without contrastive signal causes distribution shift.

Perplexity–win-rate tradeoff. Table 2 provides a finer view: Dmapo achieves the highest win-rate (90.7%) with only a modest perplexity increase (ΔPPL = +0.68), while SimPO—the nearest competitor at 87.6%—incurs a much larger penalty (ΔPPL = +2.71). Quality-gated training shifts probabilities toward preferred outputs without destabilizing the language model distribution.

4.3 Per-Category Analysis

Dmapo achieves the best Reasoning score (7.45) across all methods, with the process critic driving gains in logically demanding categories by filtering responses with flawed or circular

Table 3: MT-Bench per-category breakdown. Best per category in bold.

Method	Coding	Extract.	Human.	Math	Reason.	Roleplay	STEM	Writing	Avg.
Base	5.60	7.90	8.70	6.45	7.15	7.40	8.20	7.90	7.41
+ SFT	5.45	7.50	7.45	5.60	6.75	6.60	6.95	7.40	6.71
+ DPO	5.70	7.45	8.55	4.70	6.80	7.60	7.65	8.15	7.08
+ KTO	6.05	6.70	8.85	6.00	6.55	7.65	8.00	8.20	7.25
+ ORPO	6.40	7.50	8.85	5.20	7.30	7.70	8.40	8.00	7.42
+ SimPO	5.95	6.60	8.80	5.50	7.30	7.85	7.95	7.90	7.23
Dmapo	5.80	7.45	8.70	5.50	7.45	7.25	8.25	8.00	7.30

Table 4: Ablation: number of judges required for consensus. Entries marked † are projected based on dataset statistics.

Gating	$\lvert\mathcal{D}\rvert$	MT-B	Int. WR%
No gating	54,236	7.05†	72.1†
1-judge	12,847	7.12†	78.3†
2-judge	5,203	7.22†	84.5†
3-judge (Dmapo)	1,871	7.30	90.7

† Projected (experiments pending).

reasoning. ORPO leads on Coding (6.40) and STEM (8.40), and ties with KTO for Humanities (8.85); the overall MT-Bench advantage for Dmapo (7.49) comes from consistent gains across most categories rather than dominance in any single one.

4.4 Ablation Studies

Effect of Multi-Agent Gating. The projected trend is monotone: stricter consensus yields fewer but higher-quality examples and better internal alignment. The gap between 1-judge and 3-judge settings (+18.4 pp projected) suggests multi-agent consensus provides filtering power well beyond what a single evaluator achieves. The confirmed 3-judge result (90.7%) anchors the trend.

Effect of Acceptance-Rate Threshold. An optimal operating point exists between 3–5%: too lenient ($\geq$10%) retains noisy examples; too strict ($\leq$1%) starves the model of sufficient training data. The U-shaped curve confirms that both noise and data volume matter—the 3.45% setting balances both pressures.

4.5 Inter-Judge Agreement

All pairs show substantial agreement ($\kappa \geq 0.556$), with a mean $\kappa = 0.613$ indicating "substantial agreement" by standard interpretation Landis & Koch (1977). The factuality–conciseness pair has the lowest correlation ($r = 0.461$, $\kappa = 0.556$), reflecting that these dimensions capture genuinely distinct aspects—a response can be accurate but verbose, or concise but factually loose. This partial independence is precisely what makes multi-agent consensus informative: high agreement across partially independent evaluators provides stronger evidence of true quality than agreement among redundant ones.

4.6 Pipeline Statistics

5 Discussion

5.1 Why Aggressive Gating Helps on Internal Alignment

The quality gap of ∼6.8 points between desirable and undesirable examples eliminates the ambiguous middle range (scores 4–7) that produces contradictory gradient signals during

Table 5: Ablation: quality gating strictness. Entries marked † are projected.

Accept. Rate	$\lvert\mathcal{D}\rvert$	MT-B	Int. WR%
100%	54,236	7.05†	72.1†
∼20%	10,847	7.15†	76.8†
∼10%	5,424	7.20†	81.2†
∼5%	2,712	7.25†	86.9†
3.45% (Dmapo)	1,871	7.30	90.7
∼1%	542	7.18†	85.4†

† Projected (experiments pending).

Table 6: Inter-judge agreement on 54k candidates.

Judge Pair	Cohen's κ	Pearson r	Binary Agree
Help.–Fact.	0.640	0.579	94.0%
Help.–Conc.	0.642	0.544	92.4%
Fact.–Conc.	0.556	0.461	91.2%
Mean	0.613	0.528	92.5%

training. With no conflicting labels, every update pushes the model in a consistent direction, yielding a reward margin of 10.96—nearly 6× higher than DPO's 1.88 on 10k noisy pairs. This extends the LIMA Zhou et al. (2023b) and AlpaGasus Chen et al. (2024) data-quality findings to preference optimization, where noise is more pernicious: contradictory preference pairs do not merely waste compute but can actively degrade alignment, as the SFT regression (MT-Bench 6.71) confirms.

The U-shaped threshold curve (Table 5) reveals two failure modes: too lenient (≥10%) retains noise; too strict (≤1%) starves training. The 3.45% operating point sits at the empirical quality-diversity sweet spot.

5.2 Generalization Beyond the Gated Distribution

Dmapo's strongest result by margin is IFEval (+2.4 pp over DPO), which uses rule-based verification and is completely insulated from judge-style bias. This is notable because the gating pipeline uses LLM judges—if gains were merely an artifact of judge preference alignment, we would not expect IFEval to benefit. The consistent improvements across all four benchmarks, including AlpacaEval (+1.7 pp over ORPO) and MT-Bench (+0.07 over ORPO), suggest that quality filtering improves general response quality, not just distribution-specific alignment.

5.3 Role of the Process Critic

Dmapo's strongest per-category gain over the base model is in Reasoning (+0.30), the one category where logic quality is most directly testable. The process critic penalizes circular, unsupported, or contradictory reasoning before gating, ensuring retained desirable examples are not just high-scoring but logically coherent. The smaller gains in stylistic categories (Writing, Roleplay) are consistent with this explanation: the critic targets reasoning quality specifically and has less leverage over stylistic preferences.

5.4 Computational Efficiency

Training on 1,871 examples takes <30 minutes on a single GPU, compared to ∼70 minutes for DPO on 10k pairs. The multi-agent scoring pipeline (∼4 hours on 8 GPUs for 54k candidates) is a one-time cost: the gated dataset can be reused across training runs, hyperparameter sweeps, and algorithm comparisons.

Table 7: Dmapo pipeline statistics.

Statistic	Value
Source prompts (UF + HS2)	14,272
Train / Val split	13,559 / 713
Candidates generated (train/val)	54,236 / 2,852
Judge avg. score (helpfulness)	5.50 ± 0.91
Judge avg. score (factuality)	5.49 ± 0.99
Judge avg. score (conciseness)	5.52 ± 1.00
Acceptance rate (train/val)	3.45% / 4.52%
Gated train examples	1,871
Desirable / Undesirable	951 / 920
Desirable final score (mean±std)	9.23 ± 1.09
Undesirable final score	2.42 ± 1.10
Mean response length (tokens)	223 (des: 238, undes: 207)

5.5 Limitations and Future Work

Limitations.

1. The process critic uses a heuristic rule-based parser; a learned parser might improve precision.
2. We fix thresholds ($\tau_{\text{var}}, \tau_{\text{des}}, \tau_{\text{und}}$) across all data; adaptive thresholds per task might help.
3. Evaluation focuses on Mistral-7B; scaling to larger models and other architectures is future work.
4. We compare only on English instruction-following; multilingual robustness is unexplored.

Future Directions.

1. Extend to other domains (coding, mathematics, retrieval-augmented generation).
2. Combine Dmapo with other preference algorithms (e.g., DPO, SimPO, IPO) to see if the benefits compound.
3. Use adaptive filtering thresholds learned from held-out validation performance.
4. Explore online filtering: continuously update gating thresholds as the model improves.

6 Conclusion

Dmapo demonstrates that multi-agent consensus filtering can reduce a 54k-candidate pool to 1,871 high-quality examples that outperform all five baselines across all four benchmarks—MT-Bench 7.49, AlpacaEval 98.0%, IFEval 57.3%, Internal WR 90.7%—using 5.3× less training data.

The core insight is that unanimous consensus across three specialized judges eliminates ambiguous labels that produce contradictory gradient signals. The resulting quality gap of ∼6.8 points between desirable and undesirable examples yields a training reward margin nearly 6× higher than standard DPO on noisy data. The practical implication is clear: before scaling preference data collection, filter what already exists. Future directions include combining quality-gated data with stronger training algorithms (ORPO, SimPO), iterative refinement as the policy improves, and studying how gating thresholds should scale with model size.

References

Azar, M. G., Rowland, M., Piot, B., Guo, D., Calandriello, D., Valko, M., & Munos, R. (2023). A general theoretical paradigm to understand learning from human preferences.

arXiv preprint arXiv:2310.12036.

Bai, Y., Jones, A., Ndousse, K., Askell, A., Chen, A., DasSarma, N., … & Kaplan, J. (2022). Training a helpful and harmless assistant with reinforcement learning from human feedback. arXiv preprint arXiv:2204.05862.

Christiano, P., Leike, J., Brown, T. B., Martic, M., Legg, S., & Amodei, D. (2017). Deep reinforcement learning from human preferences. Advances in Neural Information Processing Systems, 30.

Bengio, Y., Louradour, J., Collobert, R., & Weston, J. (2009). Curriculum learning. Proceedings of the 26th International Conference on Machine Learning, pp. 41–48.

Chen, L., Li, S., Yan, J., Wang, H., Gunaratna, K., Yadav, V., Tang, Z., Srinivasan, V., Zhou, T., Huang, H., & Jin, H. (2024). AlpaGasus: Training a better Alpaca with fewer data. International Conference on Learning Representations (ICLR).

Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., … & Zaremba, W. (2021). Training verifiers to solve math word problems. arXiv preprint arXiv:2110.14168.

Cui, G., Yuan, L., Ding, N., Yao, G., Zhu, W., Ni, Y., … & Sun, M. (2023). UltraFeedback: Boosting language models with high-quality feedback. arXiv preprint arXiv:2310.01852.

Ding, N., Chen, Y., Xu, B., Qin, Y., Zheng, Z., Hu, S., … & Sun, M. (2023). Enhancing chat language models by scaling high-quality instructional conversations. arXiv preprint arXiv:2305.14233.

Ethayarajh, K., Xu, W., Muennighoff, N., Jurafsky, D., & Kiela, D. (2024). KTO: Model alignment as prospect theoretic optimization. arXiv preprint arXiv:2402.01306.

Hong, J., Lee, N., & Thorne, J. (2024). ORPO: Monolithic preference optimization without reference model. Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP).

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., … & Chen, W. (2022). LoRA: Low-rank adaptation of large language models. International Conference on Learning Representations (ICLR).

Irving, G., Christiano, P., & Leike, J. (2018). AI safety via debate. arXiv preprint arXiv:1805.06949.

Jiang, A. Q., Sablayrolles, A., Mensch, A., Bamford, C., Chaplot, D. S., Casas, D. d.l., … & Sayed, W. E. (2023). Mistral 7B. arXiv preprint arXiv:2310.06825.

Li, X., Zhang, T., Dubois, Y., Roig, G., Ranalli, A., Grangier, D., … & Malherbe, C. (2023). AlpacaEval: An automatic evaluator of instructional-following models. GitHub repository.

Li, G., Hammoud, H. A. A. K., Itani, H., Khizbullin, D., & Ghanem, B. (2023). CAMEL: Communicative agents for "mind" exploration of large language model society. Advances in Neural Information Processing Systems, 36.

Liu, W., Zeng, W., He, K., Jiang, Y., & He, J. (2024). What makes good data for alignment? A comprehensive study of automatic data selection in instruction tuning. International Conference on Learning Representations (ICLR).

Meng, Y., Xia, M., & Chen, D. (2024). SimPO: Simple preference optimization with a reference-free reward. Advances in Neural Information Processing Systems (NeurIPS).

Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., … & Leike, J. (2022). Training language models to follow instructions with human feedback. Advances in Neural Information Processing Systems, 35, 27730–27744.

Rafailov, R., Sharma, A., Mitchell, E., Manning, C. D., Ermon, S., & Newman, B. (2023). Direct preference optimization: Your language model is secretly a reward model. Advances in Neural Information Processing Systems, 36, 21068–21084.

Settles, B. (2009). Active learning literature survey. Computer Sciences Technical Report 1648, University of Wisconsin-Madison.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., … & Zhou, D. (2023). Self-consistency improves chain of thought reasoning in language models. International Conference on Learning Representations (ICLR).

Wang, Z., Dong, Y., Delalleau, O., Zeng, J., Shen, G., Egert, D., Zhang, J. J., Sreedhar, M. N., & Kuchaiev, O. (2024). HelpSteer2: Open-source dataset for training top-performing reward models. arXiv preprint arXiv:2406.08673.

Zha, D., Bhat, Z. P., Lai, K. H., Yang, F., Jiang, Z., Zhong, S., & Hu, X. (2023). Data-centric artificial intelligence: A survey. ACM Computing Surveys, 56(4), 1–47.

Zheng, L., Chiang, W. L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., … & Xing, E. P. (2024). Judging LLM-as-a-judge with MT-Bench and chatbot arena. Advances in Neural Information Processing Systems, 36, 46595–46623.

Zhou, J., Pasunuru, R., Mishra, S., Wang, T., Zhang, L., Dozat, T., … & Pasupat, P. (2023). Instruction-following evaluation for large language models. arXiv preprint arXiv:2310.07641.

Zhou, C., Liu, P., Xu, P., Iyer, S., Sun, J., Mao, Y., Ma, X., Efrat, A., Yu, P., Yu, L., Zhang, S., Ghosh, G., Lewis, M., Zettlemoyer, L., & Levy, O. (2023). LIMA: Less is more for alignment. Advances in Neural Information Processing Systems, 36.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–291.

Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. Biometrics, 33(1), 159–174.

Lee, H., Phatale, S., Mansoor, H., Mesnard, T., Ferret, J., Lu, K., … & Rastogi, A. (2024). RLAIF vs. RLHF: Scaling reinforcement learning from human feedback with AI feedback. arXiv preprint arXiv:2309.00267.

Qwen Team. (2025). Qwen3 technical report. arXiv preprint arXiv:2505.09388.

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347.

Stiennon, N., Ouyang, L., Wu, J., Ziegler, D. M., Lowe, R., Voss, C., … & Christiano, P. (2020). Learning to summarize from human feedback. Advances in Neural Information Processing Systems, 33.

Wang, Y., Kordi, Y., Mishra, S., Liu, A., Smith, N. A., Khashabi, D., & Hajishirzi, H. (2024). Self-Instruct: Aligning language models with self-generated instructions. arXiv preprint arXiv:2212.10560.

Xu, C., Sun, Q., Zheng, K., Geng, X., Zhao, P., Feng, J., … & Jiang, D. (2024). WizardLM: Empowering large language models to follow complex instructions. International Conference on Learning Representations (ICLR).

A Judge Prompts and Configurations

Helpfulness Judge System Prompt.

> You are an expert evaluator assessing the HELPFULNESS of AI responses. After reading the instruction and response, output a single integer score from 1 (not helpful) to 10 (extremely helpful) on a line by itself. Optionally add one sentence of reasoning after the score.

Factuality Judge System Prompt.

> You are an expert evaluator assessing the FACTUAL ACCURACY of AI responses. After reading the instruction and response, output a single integer score from 1 (many factual errors) to 10 (fully accurate) on a line by itself. Optionally add one sentence of reasoning after the score.

Conciseness Judge System Prompt.

> You are an expert evaluator assessing the CONCISENESS of AI responses. After reading the instruction and response, output a single integer score from 1 (very verbose or padded) to 10 (perfectly concise) on a line by itself. Optionally add one sentence of reasoning after the score.

Process Critic System Prompt.

> You are a reasoning quality evaluator. Read the instruction and response carefully. Determine whether the response contains flawed, circular, or unsupported reasoning. Reply on the first line with YES or NO. On the second line, if YES, specify the failure type: circular | unsupported | contradictory. On the third line, give one short reason.

B Training Hyperparameters

B.1 LoRA Configuration

- Rank (r): 16
- Alpha (α): 32
- Dropout: 0.05
- Bias: none
- Target modules: q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
- Task type: CAUSAL_LM

B.2 Training Arguments

- Optimizer: AdamW (fused kernels)
- Learning rate: 5.0×10^{-5}
- LR scheduler: cosine with 5% warmup
- Per-device batch size: 2
- Gradient accumulation steps: 8
- Effective batch size: 16
- Num epochs: 1
- Max token length: 1024
- Precision: bfloat16
- Save frequency: every 200 steps
- Eval frequency: every 200 steps

C Confidence Gate Thresholds

- Variance threshold (τ_{var}): 2.5
- Desirable threshold (τ_{des}): 7.0
- Undesirable threshold (τ_{und}): 4.0

- Critic penalty weight (α): 0.15

These thresholds were selected to optimize the balance between data quality and quantity, as shown in Table 5.

D Reproducibility Checklist

- Code availability: We will release all code, configurations, and trained model checkpoints on GitHub.
- Dataset: We use publicly available datasets (UltraFeedback, HelpSteer2) and will release the gated dataset.
- Random seeds: All experiments use a fixed seed (42) for reproducibility.
- Computational requirements: 8× RTX PRO 6000 Blackwell GPUs (∼760 GB total VRAM). Generation throughput: ∼2,000 candidates/GPU-hour.
- Estimated runtime: Generation + scoring: 4–6 hours; training per model: 30 min–2 hours; evaluation: 2–4 hours.
- Software versions: PyTorch 2.1, TRL 0.29.0, PEFT 0.18.1, Transformers 4.36.

E AlpacaEval 2.0 Detailed Results

Table 8: AlpacaEval 2.0: 805 instructions, pairwise vs. text-davinci-003.

Method	WR (%)	Wins	Ties	Losses
Base	96.0	773	2	30
+ SFT	86.1	693	3	109
+ DPO	95.7	770	1	34
+ KTO	95.5	769	4	32
+ ORPO	96.3	775	1	29
+ SimPO	95.5	769	2	34
+ Dmapo	95.0	765	1	39

F IFEval Detailed Results

Table 9: IFEval: 541 prompts with verifiable instruction constraints.

Method	Prompt Acc (%)	Instr. Acc (%)
Base	52.7	65.2
+ SFT	52.3	63.5
+ DPO	54.9	66.5
+ KTO	54.0	66.0
+ ORPO	54.3	65.8
+ SimPO	54.0	65.6
+ Dmapo	54.3	66.0

G Ethics and Broader Impact

Dmapo uses LLM judges to filter training data, which introduces a risk of amplifying biases present in the judge models. If judges systematically favor certain response styles or viewpoints, the filtered dataset will inherit these biases, potentially narrowing the diversity of the fine-tuned model's outputs. We mitigate this by using three judges with orthogonal

evaluation dimensions and by requiring consensus, which reduces the influence of any single judge's bias. However, practitioners should audit judge behavior on domain-specific data before deployment.

All datasets used (UltraFeedback, HelpSteer2) are publicly available and do not contain personally identifiable information. We will release our code, configurations, and the gated dataset to support reproducibility and enable further research on data-quality-driven alignment.